

Lab 5 Bonus Report

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0 Hyperparameters and Evaluation Metrics

In this experiment, we primarily adjust three hyperparameters - Learning Rate, Batch Size and Epochs. Their definitions are clear and will not be repeated here. But when it comes to evaluation metrics, it's necessary for us to clarify their definitions.

- Accuracy: the proportion of all predictions that were correct, regardless of class. It is mathematically defined as

$$Acc = \frac{\sum_{i=1}^n TP_i}{N}.$$

Accuracy is simple to define for both multi-class and binary classification. However, defining recall (the true positive rate) is not straightforward in a multi-class scenario.

In a binary setting, predictions can be partitioned into four categories: True Positives, True Negatives, False Positives, False Negatives. But in the multi-class case, a single "negative" class no longer exists; instead, each class has its own set of TPs, FPs, TNs, and FNs relative to all other classes.

- Accordingly, recall for a specific class i is defined as the proportion of actual instances of class i that are correctly predicted:

$$Rec_i = \frac{TP_i}{TP_i + FN_i}.$$

This per-class recall can then be aggregated into an overall metric using several different methods.

- Macro Recall: The arithmetic mean of all per-class recalls. It treats every class as equally important.

$$MRec = \frac{\sum_{i=1}^n Rec_i}{n}.$$

- Weighted Recall: The average of per-class recalls, but each class is weighted by its support, i.e.

$$WRec = \sum_{i=1}^n \frac{N_i}{N} Rec_i,$$

where N_i represents the actual number of samples in class i .

Since

$$Rec_i = \frac{TP_i}{N_i}$$

and

$$N_i = TP_i + FN_i$$

Substituting Rec_i :

$$\begin{aligned} WRec &= \sum_{i=1}^n \frac{N_i}{N} Rec_i \\ &= \sum_{i=1}^n \frac{N_i}{N} \frac{TP_i}{N_i} \\ &= \sum_{i=1}^n \frac{TP_i}{N} \\ &= \frac{\sum_{i=1}^n TP_i}{N} = Acc. \end{aligned}$$

Therefore, the weighted recall is always numerically equal to accuracy. However, since we do not use precision as an evaluation metric (as in Lab 5 instructions), we will still focus on these three metrics.

1 Original Model

The original model is trained using the following hyperparameter configuration:

Table 1: Hyperparameters

Learning Rate	Batch Size	Epochs
0.001	32	10

The corresponding evaluation results are shown below:

Table 2: Evaluation Metrics (Baseline)

Accuracy	Macro Recall	Weighted Recall
73.39%	71.43%	73.39%

2 Layer Addition

In this section, the SI100BNet architecture is enhanced by introducing Batch Normalization (BN) and Dropout layers.

First, three BN layers (bn1, bn2, and bn3) are added after the three convolutional layers. The resulting performance is reported below:

Table 3: Evaluation Metrics (BN Added)

Accuracy	Macro Recall	Weighted Recall
74.54%	72.79%	74.54%

Batch Normalization normalizes the activations within each mini-batch by subtracting the batch mean and dividing by the batch standard deviation. This reduces internal covariate shift, improves training stability, and accelerates convergence.

Subsequently, a Dropout layer is added to the fully connected layer. The updated results are shown below:

Table 4: Evaluation Metrics (BN + Dropout)

Accuracy	Macro Recall	Weighted Recall
73.92%	71.18%	73.92%

Dropout randomly deactivates a fraction of neurons during training, which encourages the network to learn more robust feature representations. This mechanism effectively mitigates overfitting and improves generalization performance.

Overall, the experimental results indicate that the introduction of Batch Normalization and Dropout layers leads to improved model performance.

3 Learning Rate

We first reduce the learning rate to 0.0005 and observe the resulting performance:

Table 5: Evaluation Metrics (LR = 0.0005)

Accuracy	Macro Recall	Weighted Recall
64.04%	59.09%	64.04%

Next, the learning rate is increased to 0.002:

Table 6: Evaluation Metrics (LR = 0.002)

Accuracy	Macro Recall	Weighted Recall
74.25%	72.52%	74.25%

The learning rate determines the step size used to update model parameters during optimization. An excessively small learning rate slows con-

vergence, while an overly large value may cause unstable training or prevent convergence altogether.

4 Batch Size

We first reduce the batch size to 6:

Table 7: Evaluation Metrics (Batch Size = 6)

Accuracy	Macro Recall	Weighted Recall
74.83%	72.01%	74.83%

Then, the batch size is increased to 60:

Table 8: Evaluation Metrics (Batch Size = 60)

Accuracy	Macro Recall	Weighted Recall
71.30%	68.69%	71.30%

Batch size influences both the stability of gradient estimation and training efficiency. Smaller batch sizes introduce more stochasticity, which may enhance generalization, whereas larger batch sizes yield more stable gradients but require greater computational resources.

5 Summary

This report examines the effects of network architecture modifications and hyperparameter tuning on the performance of the SI100BNet model. Model performance is evaluated using accuracy, macro recall, and weighted recall in a multi-class classification setting.

The results show that adding Batch Normalization improves training stability and overall performance, while Dropout helps enhance generalization. Experiments on learning rate and batch size indicate that overly small

learning rates or excessively large batch sizes degrade performance, whereas moderate values lead to better convergence and accuracy.

In conclusion, appropriate architectural enhancements and careful hyperparameter selection are essential for improving model performance.